

Tertiary Infotech Academy Pte. Ltd.

UEN: 201200696W

LEARNER GUIDE

For

Blender Masterclass

Course Code: C140

Conducted by

Tertiary Infotech Academy Pte. Ltd.

UEN: 201200696W

Version 1.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	30 September 2026	Initial release of the Blender Masterclass courseware: four topics, eight connected Blender labs.	Tertiary Infotech Academy

TABLE OF CONTENTS

How to Use This Guide / Course Overview	4
Topic 1: Get Started with Blender and the Creative Brief	4
Lab 1	7
Lab 2	8
Topic 2: Develop 3D Models and Prototypes	8
Lab 3	11
Lab 4	12
Topic 3: Animate and Present the 3D Prototype	12
Lab 5	15
Lab 6	16
Topic 4: Optimise, Render and Export the Model	16
Lab 7	19
Lab 8	20
Quick Reference / Support	21

How to Use This Guide

Use this guide for the detailed click-by-click procedures. The slide deck is concept-led and intentionally does not duplicate these procedures. Work on copies of the supplied Blender assets, save incremental versions, and capture the evidence named in each lab.

Course Overview

This two-day, 15-hour course builds beginner capability in Blender: interpreting creative briefs, modelling and sculpting, materials and animation, then camera, lighting, optimisation and export.

Learning Outcomes

- LO1: Analyse a creative brief and establish a suitable polygon budget.
- LO2: Develop 3D models and prototypes in Blender.
- LO3: Animate 3D models using suitable rigging, keyframe and presentation methods.
- LO4: Optimise, light, render and export 3D models for the intended use.

Topic 1: Get Started with Blender and the Creative Brief

Analyse requirements · establish budgets · control the scene

1.1 Read the creative brief as a production contract

Turn subjective intent into testable form, function and delivery criteria.

- **Purpose:** Audience, platform and viewing distance
- **Form:** Silhouette, proportion and reference views
- **Constraints:** Time, dimensions, format and budget
- **Evidence:** Named checks, views and approval owner

Applied example:

A goldfish hero asset needs a readable silhouette, animated fins, a 500-polygon target and a GLB handoff.

Review decisions:

- Separate must-haves from preferences
- Resolve missing dimensions before modelling
- Name the final approver
- Freeze a versioned brief

1.2 Set a polygon budget that matches the use

Spend geometry where it changes silhouette, deformation or shading.

- **Target:** Mobile, web, film, still or print
- **Density:** Allocate loops to visible and moving regions
- **LOD:** Plan variants for different viewing conditions
- **Measure:** Use statistics and triangulated counts

Applied example:

A 500-polygon fish keeps loops around the eye and tail while the body uses broad, clean quads.

Review decisions:

- Define whether the limit is faces or triangles
- Reserve loops around eyes, mouth and fin roots
- Keep a budget margin for revisions
- Check after modifiers

1.3 Understand Blender workspaces and editors

Workspaces arrange editors; editors expose different views of the same data.

- **3D Viewport:** Select, transform and inspect objects
- **Outliner:** Control hierarchy, collections and visibility
- **Properties:** Edit object, data, modifier and render settings
- **Timeline:** Navigate frames and keyframes

Applied example:

The Layout workspace supports blockout; Sculpting and Shading workspaces expose specialised controls without changing the model data.

Review decisions:

- Choose a workspace for the current task
- Keep the Outliner visible on complex scenes
- Confirm active object and mode
- Save layout changes only when useful

1.4 Navigate, select and transform with intent

Reliable modelling begins with controllable views, pivots and transforms.

- **View:** Orbit, pan, zoom and frame selection
- **Select:** Object, component and hierarchy selection
- **Transform:** Move, rotate and scale on constrained axes
- **Pivot:** Median, cursor and individual origins

Applied example:

Block a fish from side and top orthographic views, then inspect perspective for volume and proportion.

Review decisions:

- Use orthographic views for proportion checks
- Constrain transformations to an axis
- Apply scale before scale-sensitive tools
- Place origins deliberately

1.5 Organise objects, data and modifiers

A clean scene makes decisions reversible and handoffs understandable.

- **Objects:** Transform containers that reference data

- **Collections:** Logical groups for visibility and export
- **Names:** Role-based, consistent object/material labels
- **Modifiers:** Ordered, non-destructive operations

Applied example:

The fish body, eyes and fins are named, collected and mirrored non-destructively before detail is added.

Review decisions:

- Name objects before duplication
- Group exportable assets in one collection
- Keep a baseline before applying modifiers
- Order Mirror before Subdivision where appropriate

Lab 1: Navigate Blender and organise a production scene

Goal: Navigate the interface, distinguish Object and Edit modes, organise objects and preserve a clean baseline.

Scenario: You are onboarding to a small studio and must turn an unfamiliar Blender file into a controlled working scene.

Output: A named, organised scene with a short evidence presentation.

Step-by-step

1. Copy STARTER.blend to a new versioned working file and open it.
2. Record the Blender version shown in Help > About Blender.
3. Identify the 3D Viewport, Outliner, Properties editor and Timeline.
4. Orbit, pan, zoom and frame the selected object.
5. Switch between perspective and orthographic views and capture both.
6. Select objects in the viewport and Outliner; confirm the active object.
7. Move, rotate and scale a copy on constrained axes.
8. Enter Edit Mode, select vertices, edges and faces, then return to Object Mode.
9. Create collections for Model, Cameras and Lights.
10. Rename objects with role-based names and remove unintended duplicates.
11. Save an incremental version and reopen it to confirm integrity.
12. Update the evidence presentation with screenshots, version and result.

Acceptance checklist

- Required editors and modes are correctly identified
- Objects and collections use meaningful names
- Transforms are intentional and recorded
- Original starter remains unchanged
- Saved file reopens without missing data

Evidence to submit

- Interface screenshot
- Perspective and orthographic comparison
- Outliner hierarchy
- Transform panel
- Saved-version confirmation

Lab folder: labs/01-navigate-blender-and-organise-a-production-scene/

Lab 2: Interpret a fish brief, budget polygons and build a blockout

Goal: Translate the brief into criteria, allocate a polygon budget and create a modifier-led blockout.

Scenario: A client needs a recognisable goldfish prototype for a lightweight interactive scene.

Output: A brief, scorecard and editable goldfish blockout within the stated budget.

Step-by-step

1. Duplicate STARTER.blend and create a Brief text object or external note.
2. Record purpose, audience, viewing distance, dimensions, format and approver.
3. Define a 500-polygon target and clarify whether it refers to faces or triangles.
4. Allocate the budget across body, eyes, fins and tail.
5. Import or display side and top references where available.
6. Block the body with a low-resolution primitive and set dimensions.
7. Add eyes and fin planes using separate, named objects.
8. Use Mirror for bilateral parts and keep modifiers unapplied.
9. Check silhouette from side, top and perspective views.
10. Open viewport statistics and record the current count.
11. Remove or redirect geometry that does not support silhouette or deformation.
12. Capture the brief, budget, views and pass/fail decision.

Acceptance checklist

- Brief contains measurable requirements
- Budget definition and allocation are explicit
- Blockout silhouette matches the concept
- Modifier workflow remains editable
- Recorded count is within target or has a justified action

Evidence to submit

- Creative brief
- Polygon allocation table
- Three key views
- Modifier stack
- Viewport statistics

Lab folder: labs/02-interpret-a-fish-brief-budget-polygons-and-build-a-blockout/

Topic 2: Develop 3D Models and Prototypes

Block out form · sculpt detail · refine topology

2.1 Translate concept art into three-dimensional form

Use multiple views and landmark proportions; do not copy a single perspective literally.

- **Views:** Front, side, top and perspective references
- **Landmarks:** Eye line, widest body, fin roots and tail
- **Volume:** Primary, secondary and tertiary forms
- **Review:** Compare silhouette before detail

Applied example:

A fish blockout begins with body mass, tail plane and eye landmarks before scales or surface detail.

Review decisions:

- Choose a dominant reference view
- Mark proportion landmarks
- Resolve contradictions explicitly
- Review at intended camera distance

2.2 Model with mesh anatomy and edge flow

Vertices, edges and faces form loops that support silhouette, shading and deformation.

- **Components:** Vertices, edges, faces and normals
- **Loops:** Continuous paths around forms and joints
- **Topology:** Quad-led structure with intentional poles
- **Normals:** Face direction and smooth shading behaviour

Applied example:

Concentric eye loops support expression while longitudinal body loops preserve a smooth fish silhouette.

Review decisions:

- Add loops only for form or deformation
- Place poles away from bending areas
- Inspect face orientation
- Prefer clean local repairs

2.3 Use Edit Mode tools as a controlled vocabulary

Extrude, inset, bevel and loop cut should each answer a form or topology need.

- **Extrude:** Extend connected form
- **Inset:** Create an inner boundary
- **Bevel:** Control edge width and highlights
- **Loop Cut:** Add resolution around a continuous loop

Applied example:

The tail grows from an extrusion, while a small bevel produces a readable highlight without excessive subdivision.

Review decisions:

- Extrude along a constrained direction
- Avoid stacked accidental extrusions
- Use bevel width consistently
- Add cuts where curvature needs support

2.4 Sculpt from broad forms to controlled detail

Sculpting is most predictable when resolution, symmetry and brush purpose are managed.

- **Brushes:** Grab, clay, smooth and crease for distinct jobs
- **Symmetry:** Preserve paired form while blocking
- **Dyntopo:** Local dynamic detail with topology cost
- **Remesh:** Redistribute resolution for sculpting

Applied example:

Grab establishes the fish belly and tail arc; Smooth removes noise; Crease defines the mouth only after proportions pass.

Review decisions:

- Lock primary forms before detail
- Choose resolution for the current stage
- Save before remesh
- Retopologise if animation demands it

2.5 Review and refine the prototype

A prototype is accepted through criteria, evidence and iteration—not appearance alone.

- **Compare:** Overlay reference and inspect key views
- **Measure:** Dimensions, transforms and polygon statistics
- **Inspect:** Normals, manifold status and modifier result
- **Decide:** Accept, repair or escalate with reasons

Applied example:

The goldfish passes silhouette and 500-polygon criteria after a local tail repair and reduction of hidden underside loops.

Review decisions:

- Preserve a baseline
- Review one failure class at a time
- Capture before/after evidence
- Record remaining limitations

Lab 3: Model and sculpt the production goldfish

Goal: Develop primary and secondary forms with suitable Edit Mode and sculpting tools.

Scenario: The approved blockout needs clearer anatomy and controlled detail without losing its low-cost structure.

Output: A recognisable goldfish model with intentional form, topology and saved milestones.

Step-by-step

1. Save the starter as a new working version and inspect symmetry.
2. Mark eye line, widest body, fin roots, mouth and tail landmarks.
3. Use Grab to establish the body, belly and tail arc.
4. Use Clay or Draw to add controlled secondary volume.
5. Use Smooth sparingly to remove noise without erasing landmarks.
6. Extrude or shape fins from simple, low-density geometry.
7. Add support loops only where silhouette or deformation needs them.
8. Inspect the model from front, side, top and perspective views.
9. Compare the current silhouette with the approved blockout.
10. Record polygon statistics and any Dyntopo or remesh decision.
11. Save a milestone before adding facial or surface detail.
12. Capture annotated before/after views and explain one modelling choice.

Acceptance checklist

- Primary and secondary forms are readable
- Symmetry is intentional
- Detail level matches the stage
- Topology plan supports later animation
- Milestone files and metrics are preserved

Evidence to submit

- Landmark view
- Sculpting before/after
- Wireframe view
- Polygon statistics
- Decision explanation

Lab folder: labs/03-model-and-sculpt-the-production-goldfish/

Lab 4: Repair topology, normals and deformation zones

Goal: Diagnose and repair the supplied mesh, then prove the revised prototype meets its criteria.

Scenario: A review finds shading defects, dense areas and weak edge flow around the tail.

Output: A repaired, review-ready mesh with a defect log and before/after evidence.

Step-by-step

1. Preserve the starter and create a defect-log table.
2. Apply a diagnostic material and enable wireframe/face orientation overlays.
3. Check dimensions, transforms and object scale.
4. Select non-manifold geometry and classify each finding.
5. Merge only confirmed duplicate vertices by a controlled distance.
6. Recalculate normals outside and inspect exceptions manually.
7. Repair internal faces, holes or accidental stacked extrusions locally.
8. Review edge flow around fin roots and the tail bend.
9. Reduce unnecessary density without changing the approved silhouette.
10. Pose or bend the tail to test deformation.
11. Rerun the original checks and record pass/fail results.
12. Save the repaired file and update the evidence presentation.

Acceptance checklist

- No unintended non-manifold geometry remains
- Normals face outward
- Silhouette remains approved
- Tail deformation holds volume
- Defect log links each issue to evidence and result

Evidence to submit

- Baseline wireframe
- Face-orientation view
- Defect log
- Deformation test
- Final statistics

Lab folder: labs/04-repair-topology-normals-and-deformation-zones/

Topic 3: Animate and Present the 3D Prototype

Rig movement · keyframe action · surface and present

3.1 Apply animation principles to readable motion

Timing, spacing, anticipation and follow-through make mechanical keyframes feel intentional.

- **Timing:** Frames allocated to an action
- **Spacing:** Distance travelled between frames
- **Arcs:** Natural curved motion paths
- **Overlap:** Fins and tail follow the body

Applied example:

A swim cycle leads with the body, then offsets the tail and fins to create overlap and a continuous arc.

Review decisions:

- Block key poses first
- Choose frame rate before animation
- Exaggerate only for readability
- Review motion at real speed

3.2 Rig with armatures and controlled weights

A rig is a deformation system whose hierarchy and weights must match intended movement.

- **Bones:** Joint chain and local axes
- **Hierarchy:** Parent-child motion relationships
- **Skinning:** Bind mesh to the armature
- **Weights:** Control vertex influence

Applied example:

A simple body-to-tail bone chain bends the fish while fin bones provide subtle independent motion.

Review decisions:

- Place joints at meaningful bend zones
- Check bone roll
- Test extreme poses
- Repair weights locally

3.3 Control keyframes, interpolation and curves

Keyframes store states; interpolation and F-curves define how the values change.

- **Keyframes:** Location, rotation, scale or property states
- **Interpolation:** Constant, linear or Bézier transitions
- **F-Curves:** Editable value-over-time graphs
- **Dope Sheet:** Timing and keyframe organisation

Applied example:

The fish translates linearly across the bowl while tail rotation uses smooth cyclic curves with offset timing.

Review decisions:

- Key only intended channels
- Use linear for constant motion
- Edit handles to remove overshoot
- Cycle only seamless channels

3.4 Build materials from shaders, UVs and textures

Materials describe surface response; textures provide mapped data; UVs define how 2D data reaches the mesh.

- **Shader:** Principled BSDF response to light
- **Channels:** Base colour, roughness, metallic and normal
- **UV:** 2D coordinates with seams and islands
- **Texture:** Image data with colour-space choices

Applied example:

The fish uses a clean body UV, colour texture, moderate roughness and a subtle normal map that holds under rotation.

Review decisions:

- Set scale before unwrapping
- Place seams in low-visibility areas
- Use Non-Color for data maps
- Test under neutral light

3.5 Present the prototype for useful feedback

A review presentation should expose intent, process, evidence and the decision required.

- **Context:** Brief and target use
- **Views:** Neutral orthographic and perspective evidence
- **Motion:** Short loop at intended speed
- **Decision:** Pass, limitation and requested approval

Applied example:

The review pack shows the concept, wireframe, UV/material view and swim cycle, then requests approval to proceed to final lighting.

Review decisions:

- Use comparable before/after views
- Show statistics and version
- State open risks
- Ask a specific review question

Lab 5: UV unwrap and surface the goldfish

Goal: Create controlled UVs, textures and a Principled BSDF material that remains readable under neutral light.

Scenario: The approved mesh needs a stable PBR material for close review and animation.

Output: A surfaced fish with validated UV layout, material channels and texture dependencies.

Step-by-step

1. Open a copied starter and apply object scale if approved.
2. Mark seams along low-visibility regions and fin boundaries.
3. Unwrap the model and inspect island orientation and spacing.
4. Apply a checker texture and identify stretching.
5. Adjust seams or UVs until density is acceptably consistent.
6. Create a named material using Principled BSDF.
7. Connect base-colour texture data with the correct colour space.
8. Connect normal or roughness data as Non-Color where supplied.
9. Set plausible roughness and specular response for a wet fish surface.
10. Test the material under a neutral studio light.
11. Pack or place texture dependencies in the lab folder.
12. Capture UV, node and rendered evidence with version information.

Acceptance checklist

- UVs have no unintended overlaps
- Checker distortion is controlled
- Material channels use correct colour spaces
- Surface reads under neutral lighting
- Texture dependencies are included

Evidence to submit

- UV layout
- Checker test
- Shader node graph
- Neutral render
- Dependency inventory

Lab folder: labs/05-uv-unwrap-and-surface-the-goldfish/

Lab 6: Rig and animate a looping swim cycle

Goal: Build a simple armature, validate deformation and animate a clean loop using controlled curves.

Scenario: The surfaced fish must perform a short, readable swim cycle for stakeholder review.

Output: A rigged fish with a looped swim animation and deformation evidence.

Step-by-step

1. Save a working copy and confirm mesh transforms.
2. Create and name a body-to-tail bone chain.
3. Check bone placement, parenting and bone roll.
4. Bind the mesh to the armature using suitable weights.
5. Pose the tail and fins at their expected extremes.
6. Repair weight leakage using Weight Paint and vertex groups.
7. Set the frame rate and intended cycle length.
8. Create body and tail key poses before in-between timing.
9. Offset fin and tail motion to create overlap.
10. Inspect interpolation and F-curves for overshoot.
11. Test the first and last frame for a seamless loop.
12. Capture rig, weights, curves and real-time playback evidence.

Acceptance checklist

- Rig hierarchy and bones are named
- Extreme poses hold volume
- No unintended vertices move
- Animation loop is seamless
- Timing and interpolation match the brief

Evidence to submit

- Armature hierarchy
- Weight-paint view
- Extreme-pose test
- F-curve view
- Loop playback or frame comparison

Lab folder: labs/06-rig-and-animate-a-looping-swim-cycle/

Topic 4: Optimise, Render and Export the Model

Frame the asset · validate quality · deliver reliably

4.1 Use cameras to control what the audience sees

Camera projection, lens, composition and depth of field must serve the communication goal.

- **Projection:** Perspective or orthographic
- **Lens:** Field of view and spatial compression
- **Composition:** Focal point, balance and negative space
- **DOF:** Focus plane and controlled blur

Applied example:

A 50 mm perspective camera presents the fish naturally; orthographic cameras remain for proportion evidence.

Review decisions:

- Choose projection for the deliverable
- Avoid lens distortion that changes review judgement
- Lock approved cameras
- Use DOF sparingly in evidence views

4.2 Light form with purpose

Light direction, size, intensity and colour reveal shape while controlling contrast and render cost.

- **Key:** Primary form-defining light
- **Fill:** Controls shadow contrast
- **Rim:** Separates silhouette from background
- **World:** Ambient environment contribution

Applied example:

A large area key reveals the body curve, a soft fill retains eye detail and a rim separates the tail.

Review decisions:

- Start with one light
- Use large sources for soft highlights
- Check colour management
- Separate presentation from diagnostic lighting

4.3 Choose a render engine and settings

Workbench, Eevee and Cycles trade speed, realism and diagnostic value.

- **Workbench:** Fast geometry and studio-light review
- **Eevee:** Real-time rasterised presentation
- **Cycles:** Path-traced lighting accuracy
- **Output:** Resolution, samples, format and colour

Applied example:

The review uses Workbench wireframes, Eevee previews and a limited-sample Cycles final render.

Review decisions:

- Use Workbench for topology evidence
- Use Eevee for fast iteration
- Use Cycles when light transport matters
- Set output before final render

4.4 Optimise the complete asset

Optimisation removes cost without breaking silhouette, deformation, shading or required data.

- **Geometry:** Delete hidden or duplicate elements
- **Transforms:** Apply scale and confirm origin
- **Data:** Remove unused materials and orphan data
- **Performance:** Measure triangles, textures and render time

Applied example:

The final fish meets its target after hidden interior faces and redundant tail loops are removed, then the animation is retested.

Review decisions:

- Optimise against a measured baseline
- Protect deformation loops
- Test after each material change
- Keep the editable master

4.5 Export and verify the delivery package

An export is complete only after round-trip validation in a clean environment.

- **GLB/glTF:** Web and real-time PBR delivery
- **FBX:** Animation and pipeline interchange
- **OBJ:** Simple static mesh interchange
- **STL:** Geometry-only fabrication workflow

Applied example:

The approved fish is exported as GLB, reopened in a clean file and checked for dimensions, materials, hierarchy and animation.

Review decisions:

- Match axes, units and scale
- Export only intended objects
- Embed or package dependencies
- Re-import and compare

Lab 7: Create bubbles, light the scene and present for review

Goal: Add a lightweight bubble/force effect, build purposeful lighting and present a review decision.

Scenario: Stakeholders need a short aquarium presentation that keeps the fish readable and demonstrates controlled secondary motion.

Output: A review scene with bubbles, three-point lighting and a decision-ready presentation.

Step-by-step

1. Copy the starter and define the review camera and duration.
2. Create or inspect a bubble emitter and keep particle counts modest.
3. Set bubble lifetime, size variation and upward movement.
4. Add a force field only if it improves controlled variation.
5. Cache or bake the simulation when stable and record the range.
6. Add a key light that reveals the fish body form.
7. Add fill to retain shadow detail and a rim for separation.
8. Check the scene under a neutral diagnostic light.
9. Create a short viewport or Eevee preview.
10. Build a review slide with brief, wireframe, material and motion evidence.
11. State known limitations and the approval decision required.
12. Record stakeholder feedback and the next owner/action.

Acceptance checklist

- Secondary motion supports rather than hides the fish
- Simulation range is reproducible
- Lighting reveals silhouette and material
- Review evidence includes version and metrics
- Feedback and decision are recorded

Evidence to submit

- Particle settings
- Simulation frame
- Lighting diagram or view
- Review slide
- Decision log

Lab folder: labs/07-create-bubbles-light-the-scene-and-present-for-review/

Lab 8: Optimise, render and export the fish-bowl capstone

Goal: Optimise the full scene, render final evidence, export the intended asset and validate the package by round trip.

Scenario: The approved animated fish must be packaged for a lightweight viewer and handed off with verification evidence.

Output: A release package containing editable source, GLB/FBX export, preview and validation manifest.

Step-by-step

1. Duplicate the latest approved file and freeze a baseline inventory.
2. Remove hidden geometry, accidental duplicates and unused materials only after review.
3. Confirm dimensions, units, origins, transforms and triangle count.
4. Retest the animation after geometry and data cleanup.
5. Set the final camera, render engine, samples, resolution and output path.
6. Render a neutral evidence image and the approved presentation image.
7. Place only intended export objects in a named collection.
8. Export GLB or FBX with documented axes, scale and animation settings.
9. Open a clean Blender file and import the exported asset.
10. Compare dimensions, hierarchy, materials, normals and animation with the source.
11. Create a manifest listing source, export, textures, previews, version and known limitations.
12. Package all dependencies and mark the release pass, conditional pass or fail.

Acceptance checklist

- Final source remains editable
- Geometry and performance targets are met
- Final renders use verified settings
- Clean round-trip import passes
- Manifest and package contain all dependencies

Evidence to submit

- Before/after statistics
- Neutral and final renders
- Export settings
- Clean re-import checks
- Release manifest

Lab folder: labs/08-optimise-render-and-export-the-fish-bowl-capstone/

Quick Reference

Blender action	Purpose
Object Mode / Edit Mode	Switch between whole-object and component editing.
N panel > Item	Inspect transforms and dimensions.
Viewport Overlays > Statistics	Read vertices, faces and triangles.
Face Orientation	Find inward-facing polygons.
Select > Select All by Trait > Non-Manifold	Locate open or ambiguous surface elements.
Ctrl+A > Scale	Apply scale only after dimension verification.
UV Editing workspace	Mark seams, unwrap and inspect distortion.
Shader Editor	Build and inspect PBR material nodes.
File > Export	Create target-specific GLB, FBX, OBJ or STL deliverables.
Scripting workspace	Review and run Blender Python scripts.

Support

Support: enquiry@tertiaryinfotech.com · +65 6100 0613